SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I M.DEVI SRILATHA(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Auto-Scaling and Load Balancing for Seasonal Traffic

A company may experience sudden increases in website traffic during seasonal sales such as Diwali, Black Friday, or New Year sales. Cloud auto-scaling and load balancing can help manage this traffic.

Auto-Scaling

Auto-scaling automatically increases or decreases the number of servers according to the traffic.

- When traffic increases, new servers/instances are automatically created.
- When traffic decreases, unnecessary instances are removed.
- This avoids wasting resources during normal periods.
- It helps the application handle a large number of users.

Load Balancing

A load balancer distributes incoming user requests among multiple servers.

For example:

Users → Load Balancer → Server 1 | Server 2 | Server 3 | Server 4

If one server becomes overloaded, requests can be redirected to other available servers.

Benefits

- Reduces server overload.
- Reduces downtime.
- Improves application response time.
- Provides better availability.
- Improves user experience.
- Uses cloud resources efficiently.

Role of Monitoring

Monitoring tools continuously check CPU usage, memory, network traffic, response time, and server health. When traffic increases beyond a defined limit, monitoring systems can trigger auto-scaling.

Real-World Example

For example, an e-commerce website such as Amazon may receive millions of users during a major sale. Auto-scaling can add more servers during peak traffic, while the load balancer distributes customers among those servers.

Additional Optimization Techniques:

- Use predictive auto-scaling based on historical sales patterns to pre-warm servers before an expected traffic surge.
- Apply queue-based load leveling so sudden order bursts are processed smoothly instead of overwhelming backend services.
- Cache frequently accessed product and pricing data to reduce repeated database load during peak sales events.

Conclusion:

Auto-scaling and load balancing help e-commerce companies handle unpredictable traffic while reducing downtime and providing a smooth shopping experience.

2. Secure Cloud Storage for Backup Data

Cloud storage is useful for organizations that need to store large amounts of backup data. Cloud providers use data redundancy, encryption, and access control to protect backup information.

Durability and Data Redundancy

Cloud providers store multiple copies of data in different storage systems or locations.

For example:

Original Data → Copy 1 → Copy 2 → Copy 3

If one storage device fails, another copy can be used.

This provides:

- High durability.
- Data availability.
- Protection against hardware failure.
- Recovery from accidental data loss.

Encryption

Encryption converts data into an unreadable form.

- **Encryption at rest** protects stored backup data.
- **Encryption in transit** protects data while it is being transferred.
- Encryption keys should be properly managed and protected.

Access Control

Access control ensures that only authorized users can access backup data.

Methods include:

- User authentication.
- Strong passwords.
- Multi-factor authentication.
- Role-based access control.
- Least-privilege permissions.

Cost Advantages

Cloud storage can be cheaper than maintaining traditional storage systems because:

- No need to purchase large physical storage systems.
- Less hardware maintenance.
- Pay only for the storage used.
- Storage can be increased when required.
- Reduced electricity and data-center costs.

Disaster Recovery Example

Suppose a company's main datacenter is damaged because of a natural disaster. The company can use its **cloud backup** to restore important databases and applications at another location.

Compliance and Retention Policies:

- Define clear data-retention schedules so backups are kept only as long as required by policy or regulation.
- Apply immutable storage or write-once policies to protect backups from accidental or malicious modification.
- Regularly test backup restoration to confirm recovery procedures work correctly, not just that backups exist.

Conclusion:

Cloud storage provides **durability, redundancy, security, scalability, and cost savings**, making it suitable for backup and disaster recovery.

3. CDN and Global Application Performance

A **Content Delivery Network (CDN)** is a group of geographically distributed servers used to deliver content to users from a location closer to them.

Without CDN:

User → Main Cloud Server → Content

With CDN:

User → Nearby Edge Server → Content

How CDN Improves Performance

CDNs store frequently accessed content such as:

- Images.
- Videos.
- Web pages.
- JavaScript and CSS files.
- Other static content.

When a user requests content, it can be delivered from a nearby **edge server** instead of a distant main server.

Benefits

- Reduces network latency.
- Improves page loading speed.
- Reduces load on the main server.
- Provides better performance during high traffic.
- Improves user experience.

Role of Edge Servers

Edge servers are located closer to users in different geographical regions. Therefore, users do not always need to communicate with a central server located far away.

For example, a user in India can receive content from an edge location in India or a nearby region instead of a server located in the US.

Global Availability

Cloud providers have datacenters and edge locations in different parts of the world. Applications can be deployed across multiple regions to provide better availability and fault tolerance.

Real-Time Streaming Example

Consider a **live sports streaming application**. Thousands or millions of users may watch the same event simultaneously.

The CDN can deliver video content from nearby edge servers, reducing latency and improving streaming quality.

Similarly, in online gaming, cloud regions and edge services can help players connect to nearby infrastructure and reduce network delay.

Additional Optimization Techniques:

- Use modern protocols such as HTTP/3 to reduce connection overhead and improve delivery speed over unreliable networks.
- Compress and optimize images and video segments before caching them at edge locations.
- Use anycast routing so user requests are automatically directed to the nearest healthy edge server.

Conclusion:

CDNs and edge servers improve **speed, availability, and user experience** for globally distributed applications.

4. Cloud Security and Compliance for Financial Institutions

Financial institutions handle sensitive information such as **bank accounts, transaction details, and customer information**. Therefore, cloud security and regulatory compliance are very important.

Regulatory Compliance

Cloud providers provide security services and compliance features that help organizations meet applicable regulations and standards.

Organizations can use:

- Compliance reports.
- Security policies.
- Audit tools.
- Identity management.
- Data encryption.
- Security monitoring.

Logging

Logging records activities performed on cloud resources.

For example, logs can record:

- Who accessed the data.
- When the data was accessed.
- What changes were made.
- Login and authentication activities.

These logs help during security investigations and audits.

Monitoring

Monitoring continuously checks cloud systems for unusual activities.

For example, if a user suddenly attempts to access a large amount of sensitive financial information, monitoring systems can generate an alert.

Encryption

Encryption protects sensitive data from unauthorized access.

- Data at rest is encrypted when stored.
- Data in transit is encrypted during communication.

Data Sovereignty

Data sovereignty means that data may need to be stored and processed according to the laws of a particular country or region.

The organization can:

- Select appropriate cloud regions.
- Restrict where data is stored.
- Use region-specific storage.
- Apply policies preventing unauthorized data movement.

Example

Consider a bank processing an online money transfer.

Customer → Banking Application → Cloud Service → Transaction Database

The transaction data can be encrypted, access can be controlled using IAM, and every transaction can be logged and monitored for suspicious activity.

Emerging Practices:

- Adopt a zero-trust security architecture so every transaction request is continuously verified, not just at login.
- Use confidential computing to protect sensitive financial data even while it is being processed in memory.
- Automate compliance scanning so configuration drift from regulatory baselines is detected and corrected quickly.

Conclusion:
Cloud providers support financial institutions through **encryption, access control, logging, monitoring, auditing, and regional data controls**, helping them protect sensitive financial information.

5. Containers, Virtual Machines and Kubernetes

Containers and virtual machines are technologies used to deploy applications, but they work differently.

Containers vs Virtual Machines

Feature	Containers	Virtual Machines
Operating System	Share host OS kernel	Each VM has its own OS
Resource Usage	Low	Higher
Startup Time	Very fast	Comparatively slower
Performance	Near-native performance	More overhead
Size	Lightweight	Usually larger
Portability	Very high	Also portable but heavier

Containers

A container packages an application along with its required libraries and dependencies.

For example:

Application + Libraries + Dependencies → Container

Containers are lightweight and can run multiple application services on the same host efficiently.

Kubernetes Orchestration

When an organization has hundreds of containers, manually managing them becomes difficult. **Kubernetes** helps automate container management.

It can:

- Deploy containers.
- Scale applications.
- Restart failed containers.
- Perform load balancing.
- Manage application updates.
- Maintain the desired number of application instances.

For example, if user traffic increases:

Low Traffic → 3 Containers

- The application receives a small number of user requests.
- Three containers are enough to handle the workload.
- This reduces resource usage and operational cost.
- The load balancer distributes incoming requests evenly among the 3 containers.
- As traffic increases, more containers can be added automatically (auto-scaling).

Example:

- 100 users visit a website.
- The load balancer sends requests to Container 1, Container 2, and Container 3.
- Each container processes part of the traffic, ensuring smooth performance without wasting resources.

High Traffic → Kubernetes → 10 Containers

After traffic decreases, unnecessary containers can be removed.

Portability

Containers can run consistently across different environments if the required container runtime and supporting services are available.

For example, the same containerized application can be deployed on:

- AWS.
- Microsoft Azure.
- Google Cloud.
- Private cloud.
- Local datacenter.

This reduces dependency on a particular cloud environment.

Microservices Example

Consider an **online shopping application**. It can be divided into different microservices:

E-Commerce Application

- User Service
- Product Service
- Order Service
- Payment Service
- Inventory Service

Each service can run in its own container.

If the **Order Service** receives heavy traffic, Kubernetes can scale only that service instead of scaling the entire application.

Additional Kubernetes Features:

- Use a service mesh (such as Istio) to manage secure service-to-service communication, retries, and traffic control between microservices.
- Package deployments with Helm charts to make application rollout and version upgrades repeatable and consistent.
- Adopt GitOps practices so infrastructure and application changes are deployed automatically from a version-controlled repository.

Conclusion:

Containers are lightweight and portable, while Kubernetes provides automated **deployment, management, scaling, and recovery**. This makes containerization suitable for modern microservices-based applications.